MVSR ENGINEERING COLLEGE

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Department of Computer Science and Engineering

Project Title: Smart Life Scheduler.

Batch-ID:DS20

Student Team Members:

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Guide: T Laxmi, Asst Prof, Dept. of CSE, MVSREC

Problem Statement: In today's fast-paced environment, individuals struggle to balance demanding schedules with maintaining proper health. Traditional task scheduling tools help users manage daily activities but fail to consider their physical and mental well-being. Similarly, existing health- monitoring applications operate independently, offering no connection between a user's workload, habits, and health conditions. As a result, users lack awareness of how factors like sleep quality, stress levels, exercise patterns, and daily tasks influence one another. This leads to poor decision-making and unhealthy routines. Therefore, there is a need for an integrated system that unifies task planning, health tracking, data analysis, and automated health reporting to support balanced lifestyle management.

Project Scope:

1. Build a mobile app for task scheduling and health monitoring.
2. Integrate APIs like Google Fit/Apple Health for activity data.
3. Use Firebase/MongoDB for data management and ML for analysis.
4. Generate automated health reports with insights and trends.

Project Objectives:

1.Provide an easy tool to plan daily tasks.
2.Track, sleep, exercise, stress and
symptoms.
3.Analyze links between tasks and health patterns.
4.Offer personalized health reports and suggestions.

Application Areas	Challenges
1. **Personal Health Tracking-**Helps users monitor sleep, stress and activity daily. 2. **Task and Routine Management-**Allows users to plan and organize their daily tasks efficiently. 3. **Fitness Monitoring-**Tracks steps, workouts and physical activity trends. 4. **Employee Wellness Programs-**Supports companies in improving employee's health and productivity. 5. **Preventive Health-**Provides early insights into unhealthy patterns through reports.	1. **Data Integration Issues-**Syncing accurate data from Google Fit/Apple Health can be complex. 2. **Privacy and Security Concerns-**Sensitive health data must be protected with strong security. 3. **Limited ML Training Data-**Health insights may be Less accurate with small datasets. 4. **User Engagement-**Users may not consistently input their tasks or symptoms. 5. **Cross-platform Performance-**Ensuring smooth functionality on both android and iOS is difficult.

Literature Survey

SL	Major Reference(s)	Finding(s)
1	Google Fit API Documentation	Provides accurate step count, activity type and heart-rate success for integrating real time health data into mobile apps.
2	Apple Health Kit API Documentation	Enables secure collection of sleep, heart rate and exercise metrics from IOS devices for personalized health monitoring.
3	Firebase Fire store Documentation	Offers scalable, real-time cloud storage for user tasks, health records and reports with built in authentication and security rules.
4	Research on Digital Wellness & Productivity Tools	Suggests that task scheduling apps reduce stress and increase productivity when combined with timely reminders and health insights.
5	WHO Health Monitoring Guidelines	Highlights the importance of tracking the daily physical activities of an individual for early detection of lifestyle-based health records.

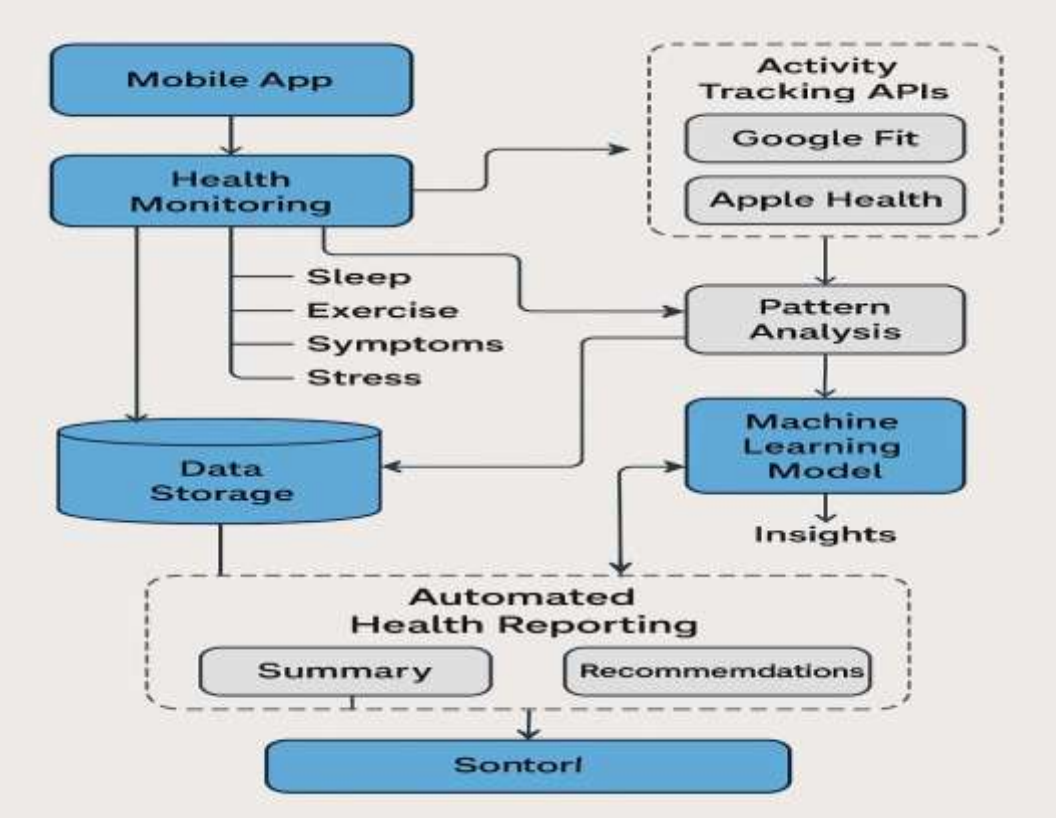
Comparison Table

SI no	Title	Paper/Reference	Methodology Used	Dataset Used
1.	Smart Life Scheduler: An AI-Based Personalized Daily Scheduling System.	Proposed System.	Machine Learning, Rule-Based Logic, AI Chat bot.	User schedules, personal preferences, lifestyle data.
2.	AI-Driven Task Scheduling and Productivity Optimization.	Related Work – 2022.	Machine Learning Algorithms.	User activity logs and task history.
3.	Health-Aware Lifestyle Recommendation System.	Related Work – 2021.	Data Analytics and Recommendation Models.	Health records and fitness tracking data.
4.	Automated Personal Planner Using Rule-Based Systems.	Related Work – 2020.	Rule-Based Expert System.	Static user schedule datasets.

Performance Metrics, Domain, Limitations & Future Scope Table:

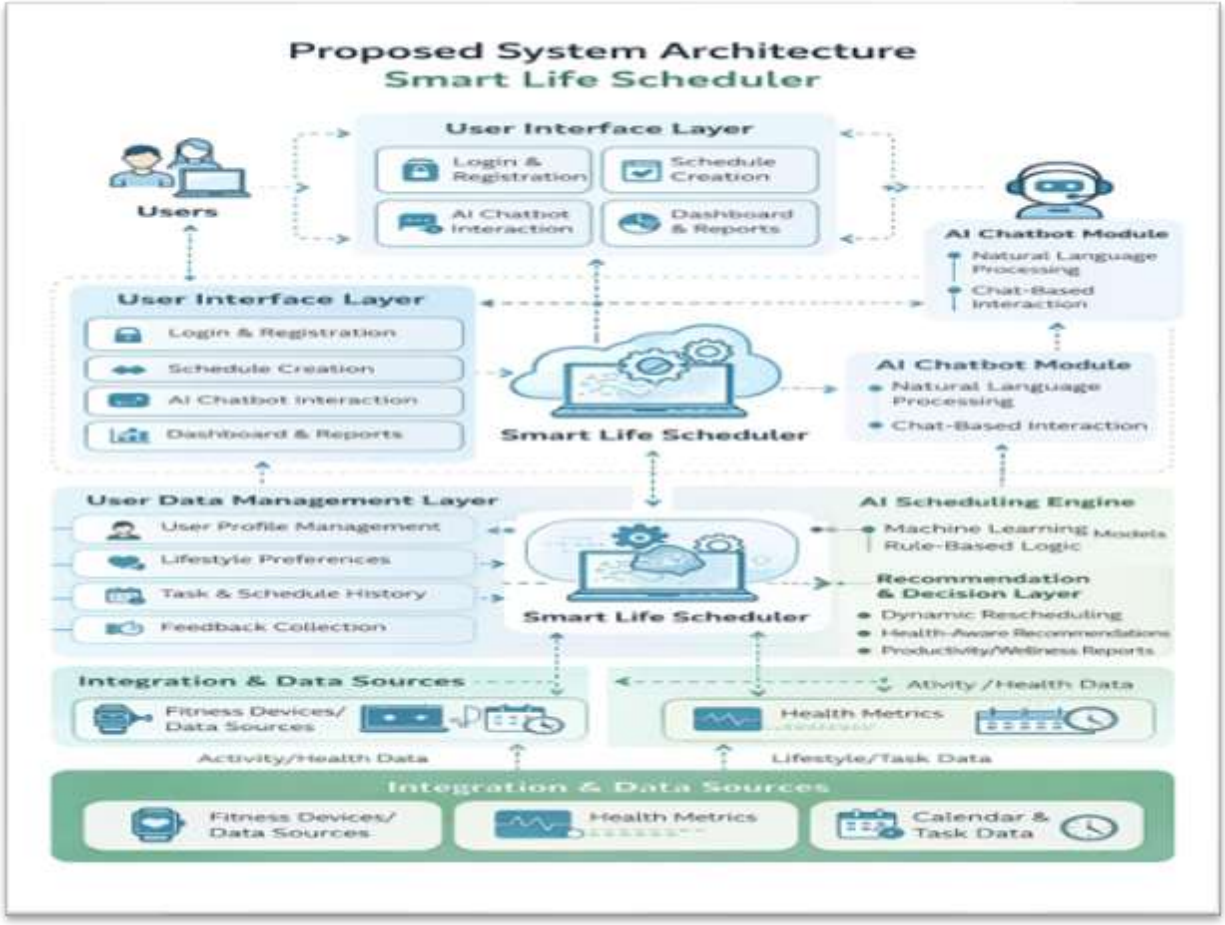
Sl. No.	Performance Metrics	Domain	Limitations / Research Gaps	Future Scope
1.	Schedule accuracy, task completion rate	AI-Based Scheduling	Does not consider health parameters	Health-aware personalized scheduling
2.	User productivity improvement	Smart Lifestyle Management	Static schedules without adaptation	Dynamic rescheduling using AI
3.	Health balance, wellness score	Health Informatics	No integration with fitness apps	Integration with wearable devices
4.	User satisfaction, system adaptability	Intelligent Systems	Rule-based and non-learning systems	ML-driven adaptive life scheduling models

Generic Architecture of the Existing Systems:



Software / Hardware requirements for development / deployment

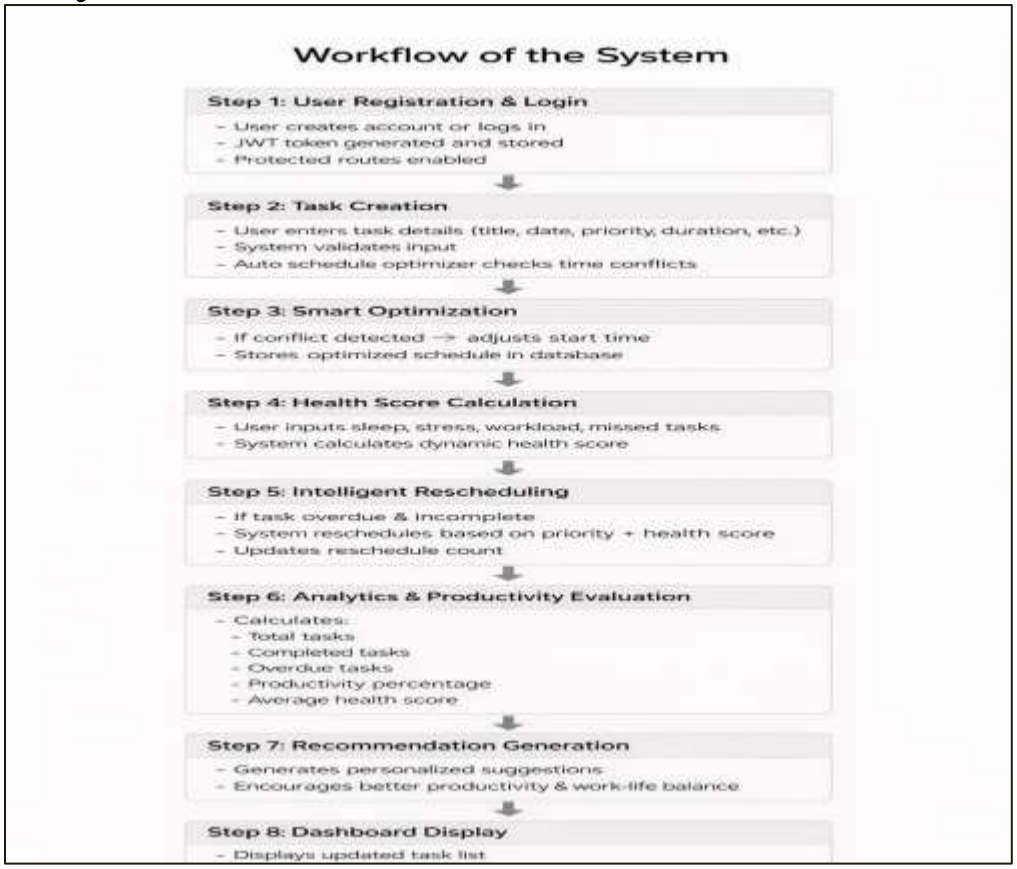
Tool / Technology	Usage
Flutter/React Native	Used to build a cross-platform mobile app with a smooth and responsive UI.
Node js/Django REST	Used to create backend API for data handling, Authentication and communication between app and database.
Firebase Fire store/MongoDB/SQL	Used to store tasks, health data, activity logs and system generated reports securely.
Python (Scikit- learn, Pandas, Numpy)	Used to perform health data analysis and run machine learning models for pattern detection and insights.
Google Fit API/Apple Health kit API	Used to fetch step count, sleep data, heart rate and physical activity directly from a user device.
Firebase Cloud Messaging/Report lab	Used for sending real time push notifications and generating downloadable Pdf health reports.


Proposed System Architecture
Smart Life Scheduler
Users
User Interface Layer
Login & Registration
Schedule Creation
AI Chatbot Interaction
Dashboard & Reports
AI Chatbot Module
Natural Language Processing
Chat-Based Interaction
User Interface Layer
Login & Registration
Schedule Creation
AI Chatbot Interaction
Dashboard & Reports
Smart Life Scheduler
AI Chatbot Module
Natural Language Processing
Chat-Based Interaction
User Data Management Layer
User Profile Management
Lifestyle Preferences
Task & Schedule History
Feedback Collection
Smart Life Scheduler
AI Scheduling Engine
Machine Learning Models
Rule-Based Logic
Recommendation & Decision Layer
Dynamic Rescheduling
Health-Aware Recommendations
Productivity/Wellness Reports
Integration & Data Sources
Fitness Devices/ Data Sources
Activity/Health Data
Health Metrics
Lifestyle/Task Data
Integration & Data Sources
Fitness Devices/ Data Sources
Health Metrics
Calendar & Task Data

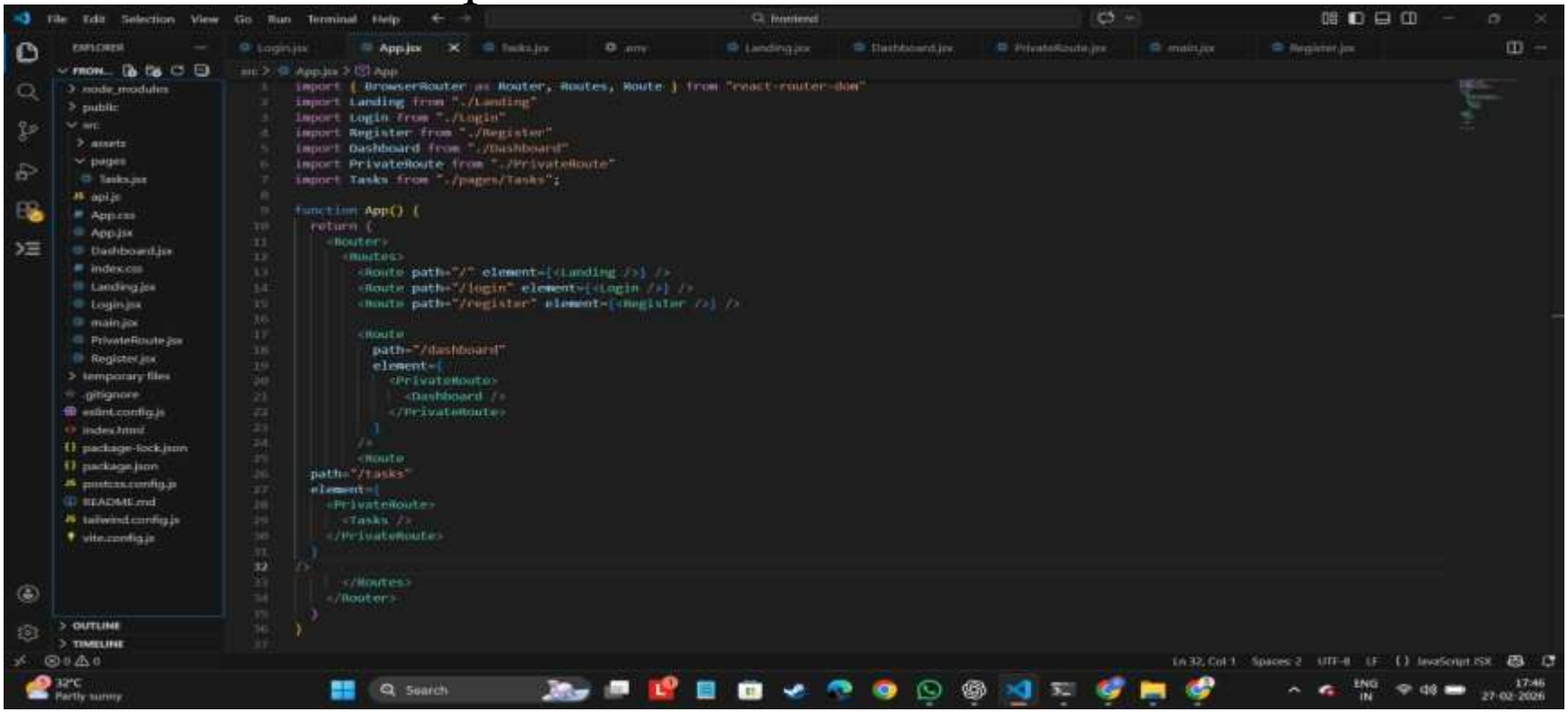
Module Description:

Name of Module	Description
User Authentication Module	Handles User Registration, Login, JWT-based authentication and secure session management using token verification and blacklist mechanism.
Task Management Module	Allows Users to create, view, update, toggle and delete tasks with full CRUD functionality.
Auto Scheduling Optimization Module	Detects scheduling conflicts between tasks. Automatically adjusts task start times to prevent overlapping and optimize time utilization.
Smart Rescheduling & Health Scoring Module	Analyzes user sleep hours, stress levels, workload, and missed tasks. Calculates a health score and intelligently reschedules overdue tasks based on user well-being.
Analytics & Productivity Module	Computes key productivity metrics such as Total tasks, Completed tasks, Overdue tasks, Average health score. Provides insights for better decision-making.
Recommendation Engine Module	Generates personalized productivity and health recommendations. Uses user performance data to suggest improvements and optimize daily routines.
Dashboard Visualization Module	Displays structured summaries of tasks and analytics.
Deployment & Cloud Integration Module	Deploys the backend application on Render. Integrates MongoDB Atlas for scalable cloud-based database management.

Workflow of the System:

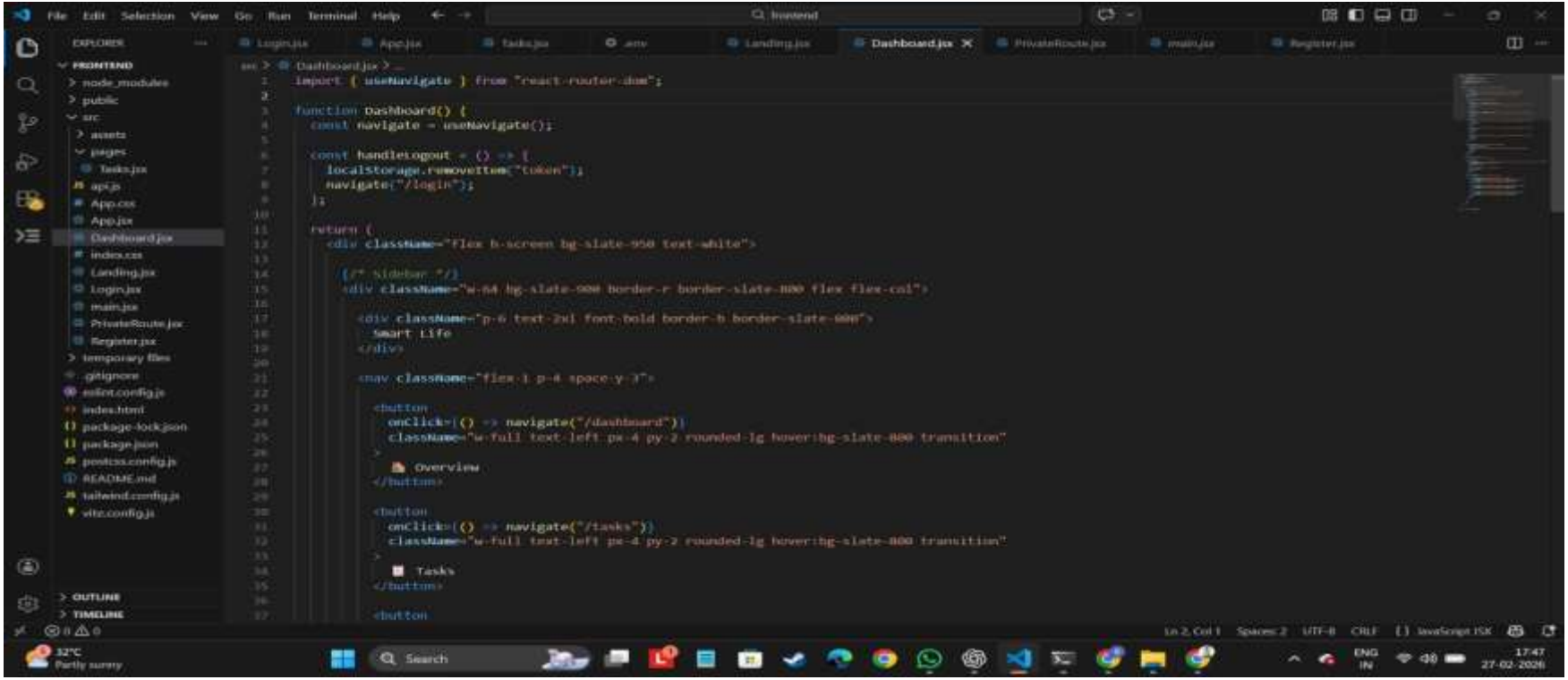


Environmental Setup:



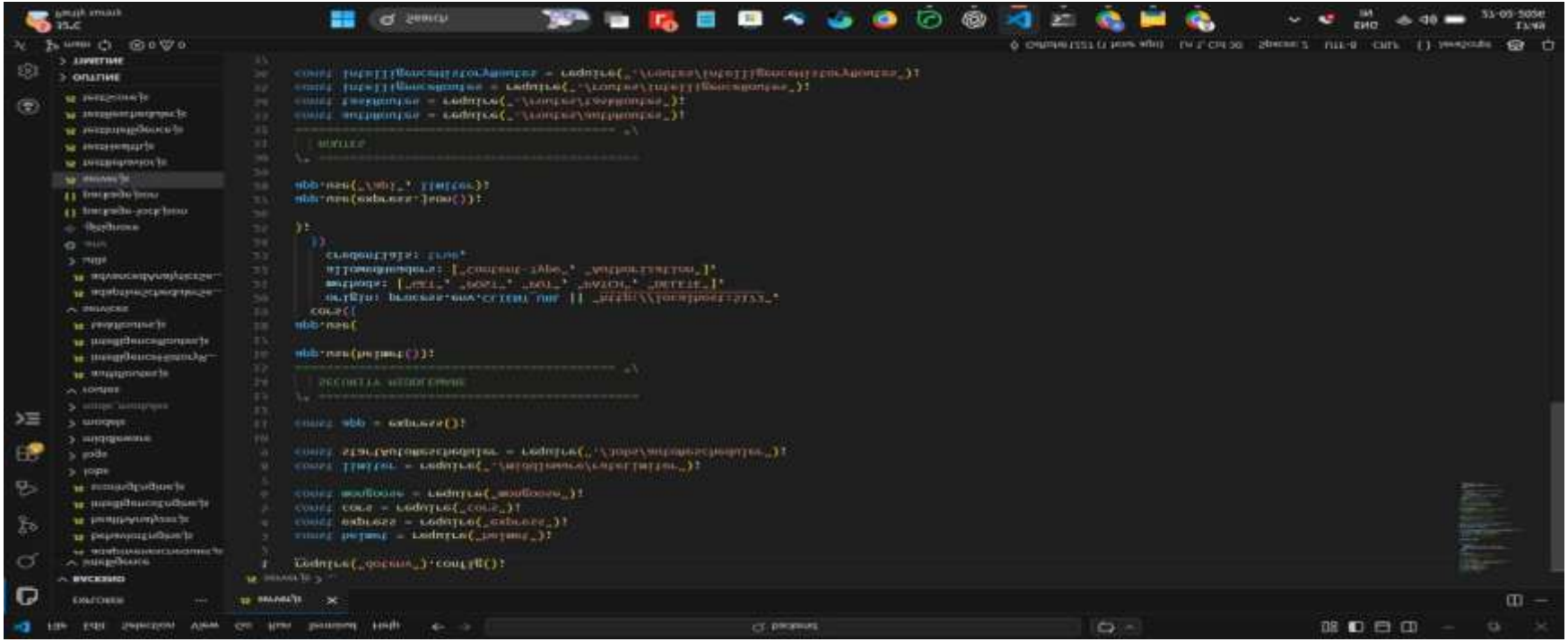
The Project is implemented using VS Code Text Editor.

Frontend:


Dashboard.jsx
import { useNavigate } from "react-router-dom";
function Dashboard() {
const navigate = useNavigate();
const handleLogout = () => {
localStorage.removeItem("token");
navigate("/login");
};
return (

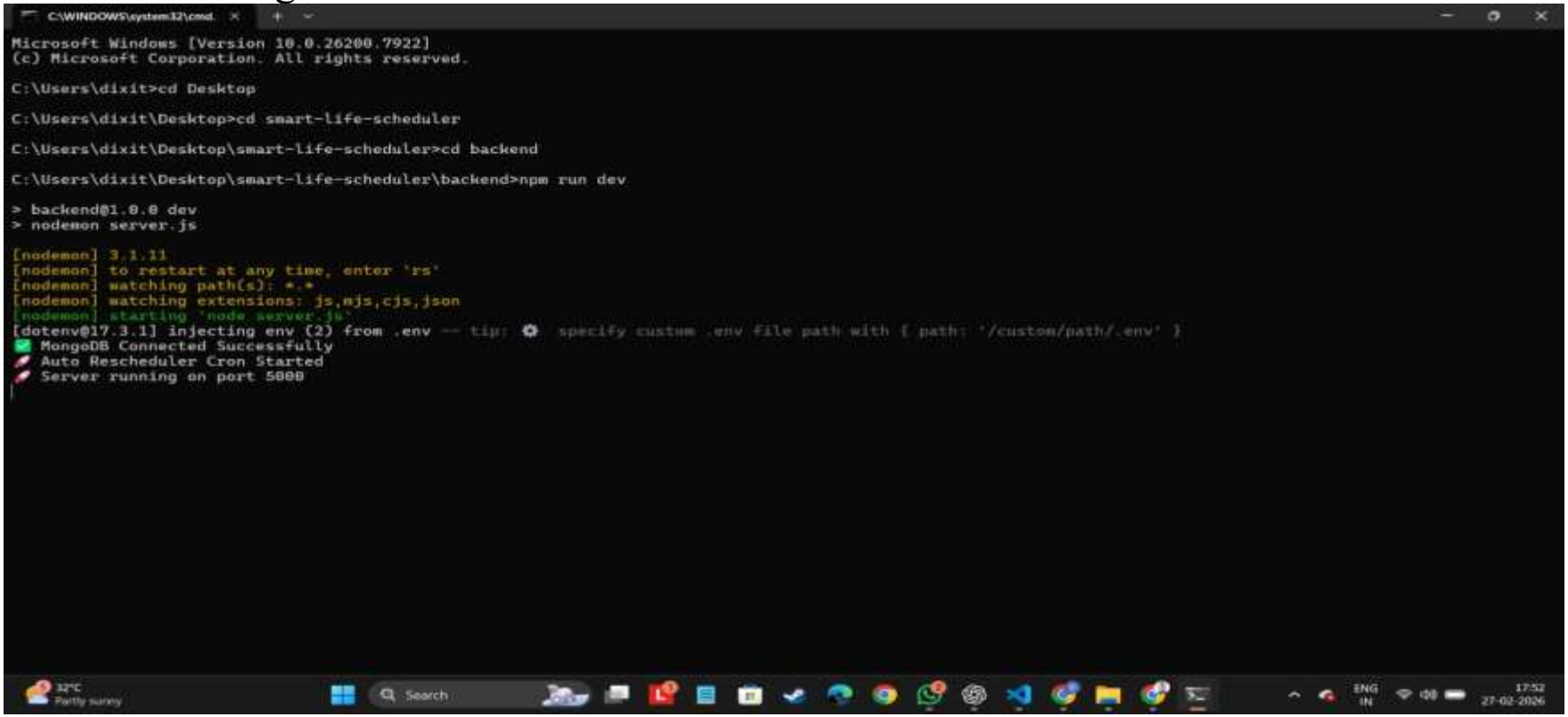
{/* Sidebar */}

Smart Life
<nav>
<button>Overview</button>
<button>Tasks</button>
<button>

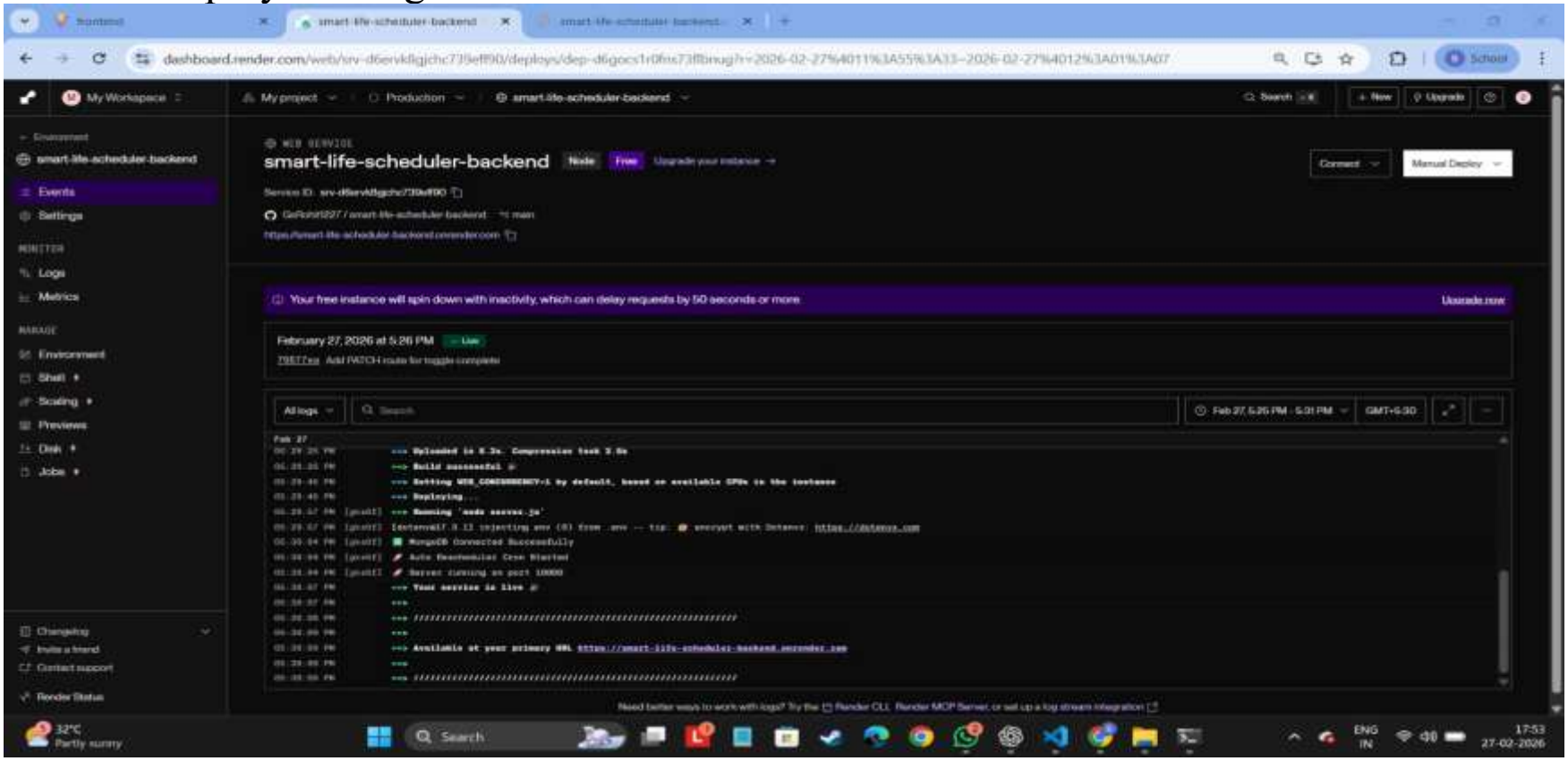


Backend:

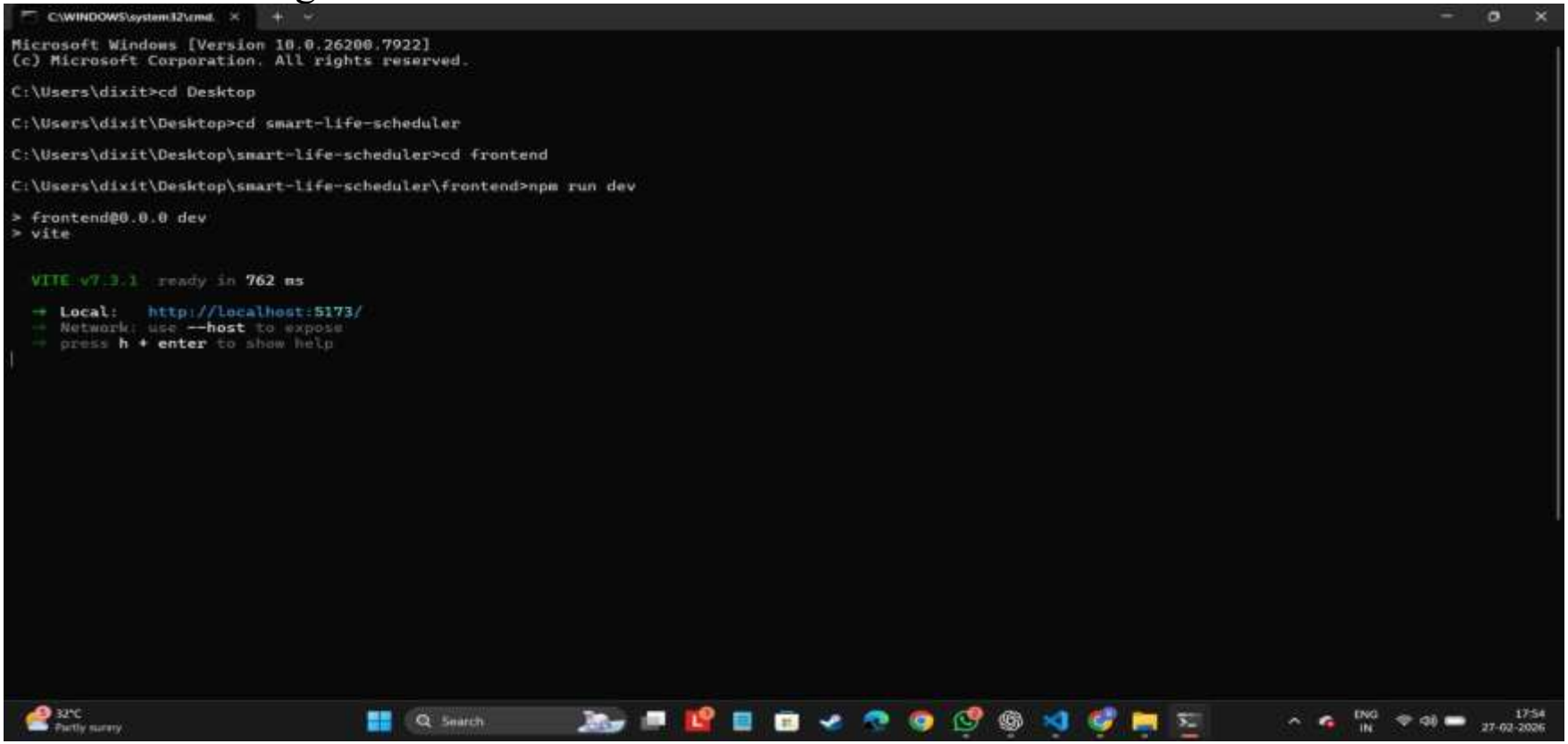
Backend running on Server:



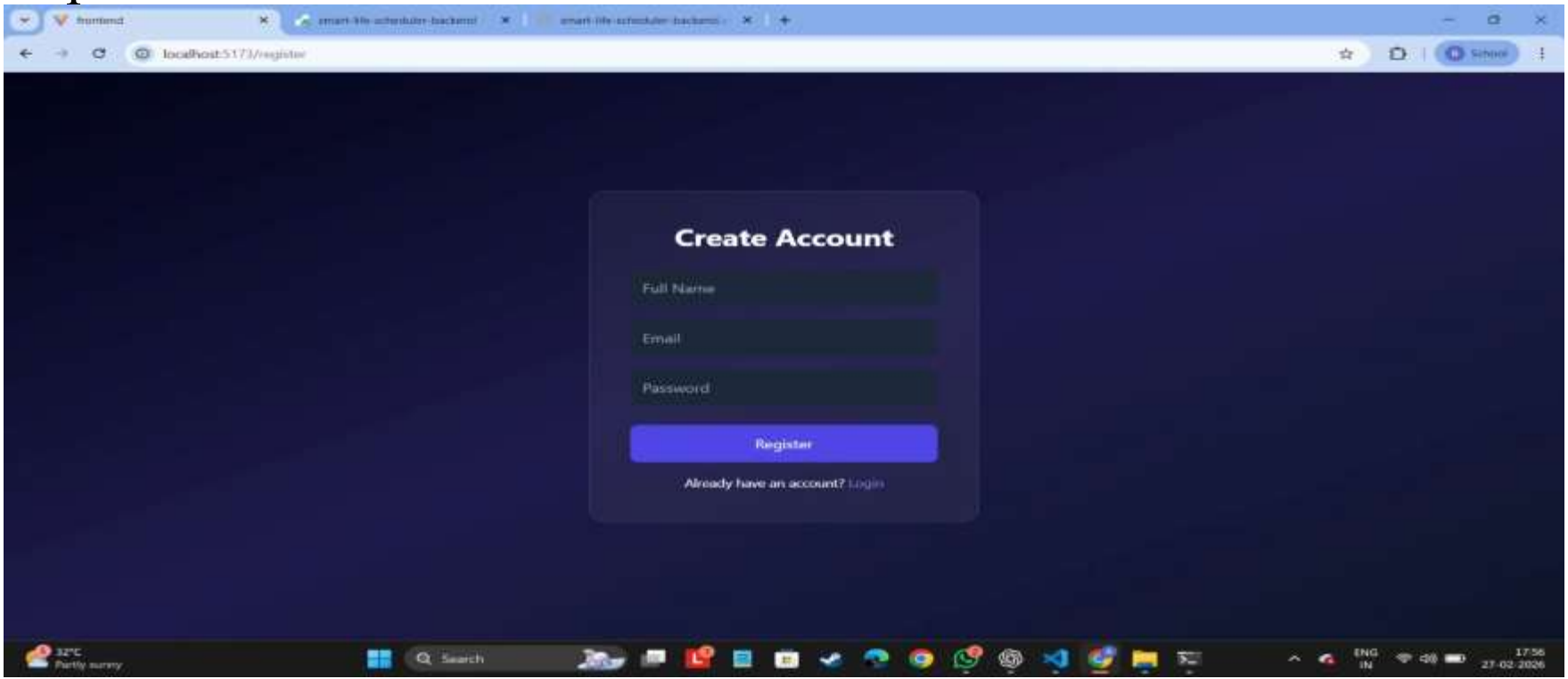
Backend deployed using render.com:



Frontend Running in local:

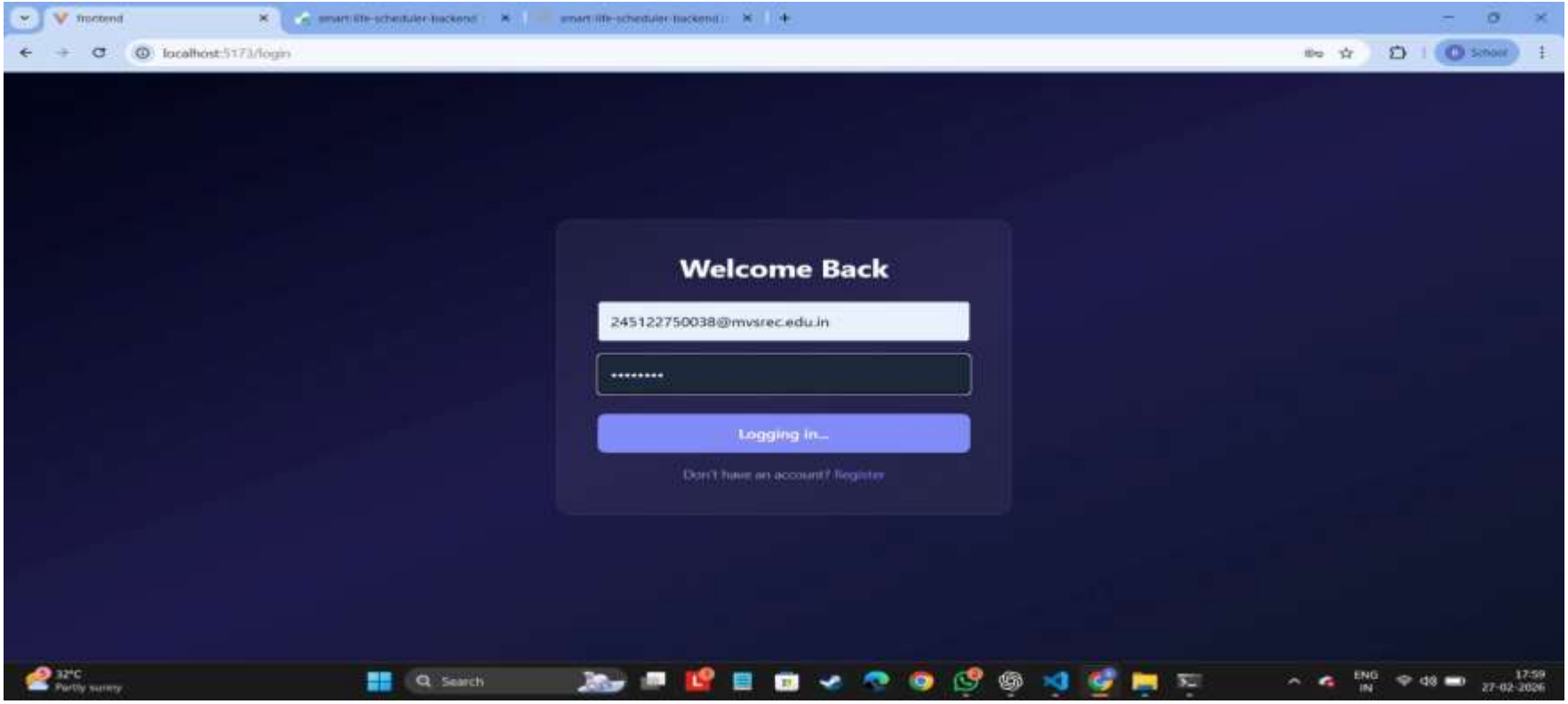


Implementation:

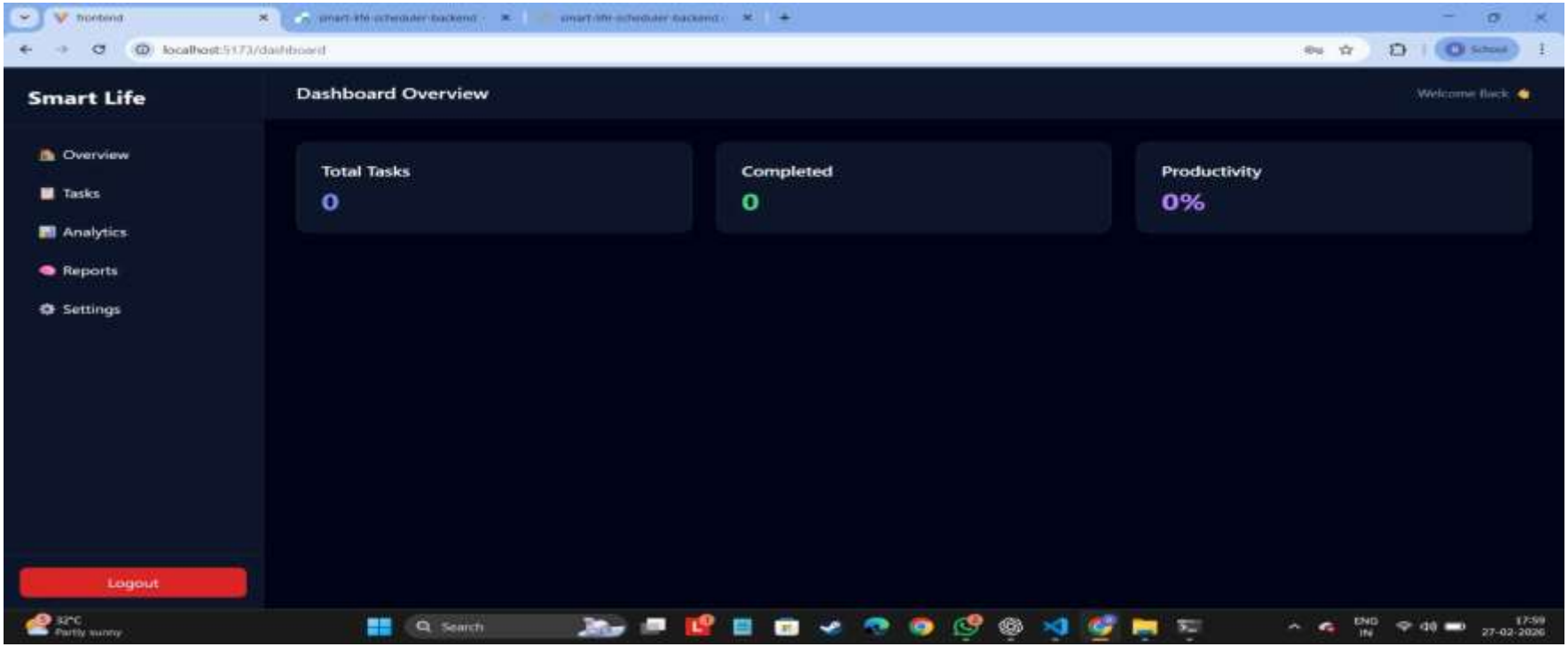


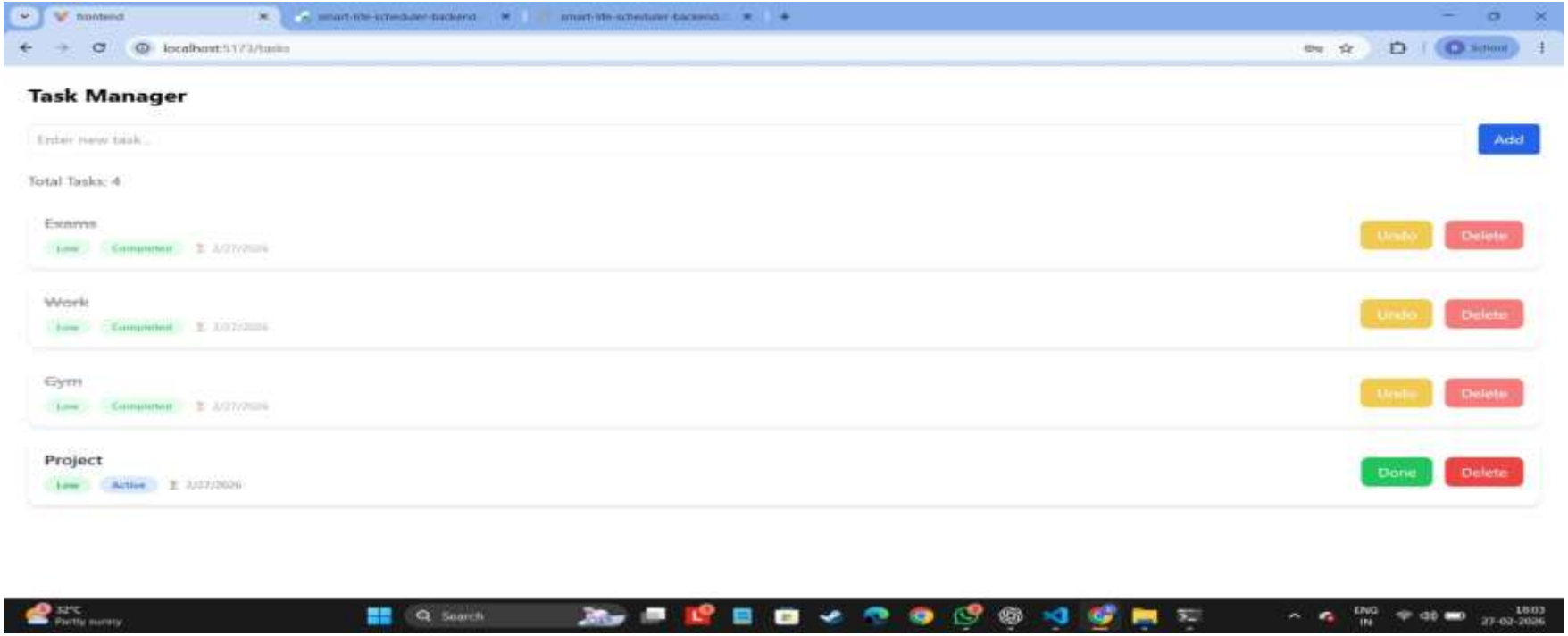
Register page:

Login page:

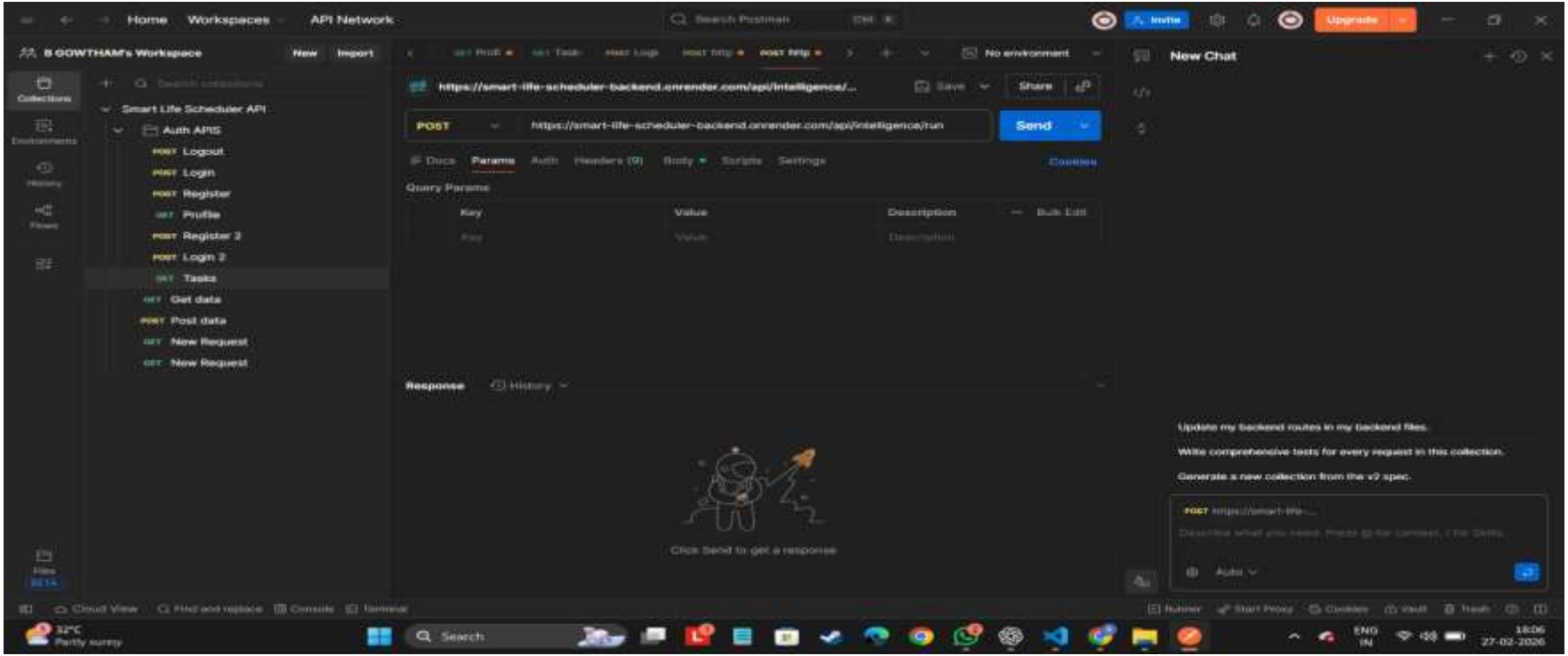

localhost:5173/login
Welcome Back
245122750038@mvsrec.edu.in
Logging in...
Don't have an account? Register

Dashboard:

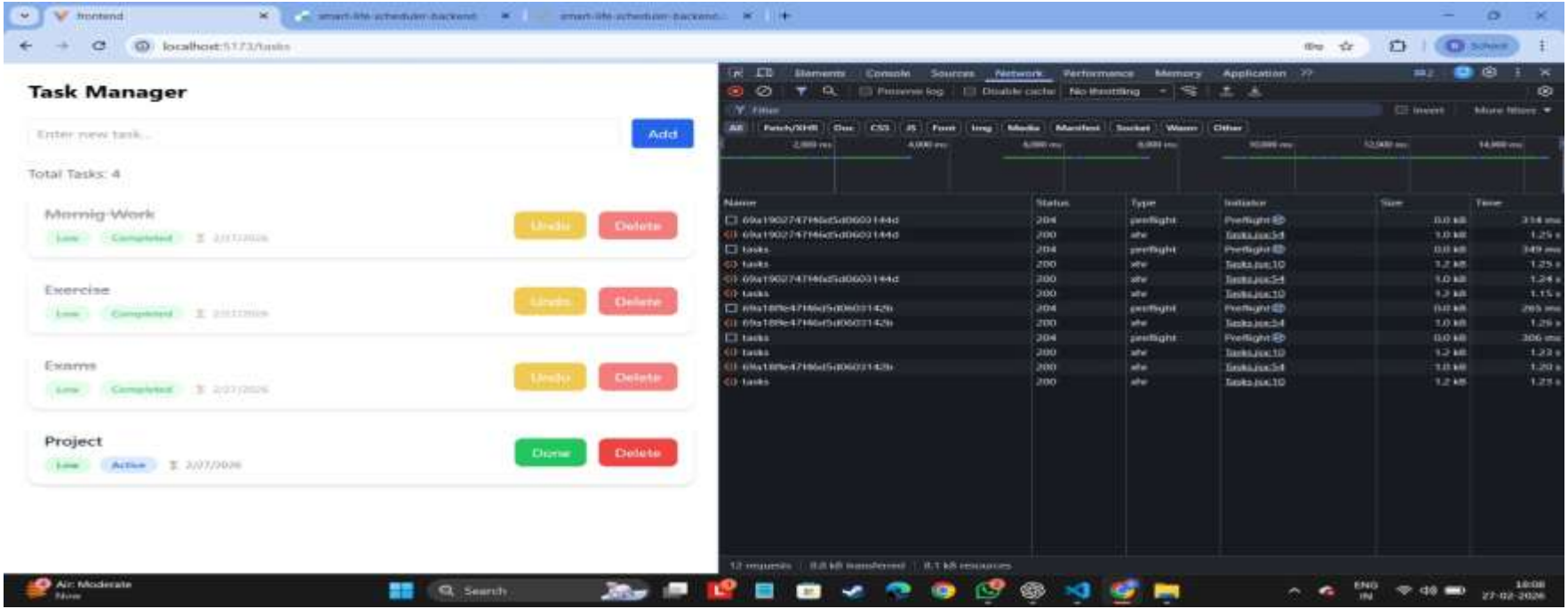

Smart Life
Dashboard Overview
Welcome Back
Overview
Tasks
Analytics
Reports
Settings
Total Tasks
0
Completed
0
Productivity
0%
Logout



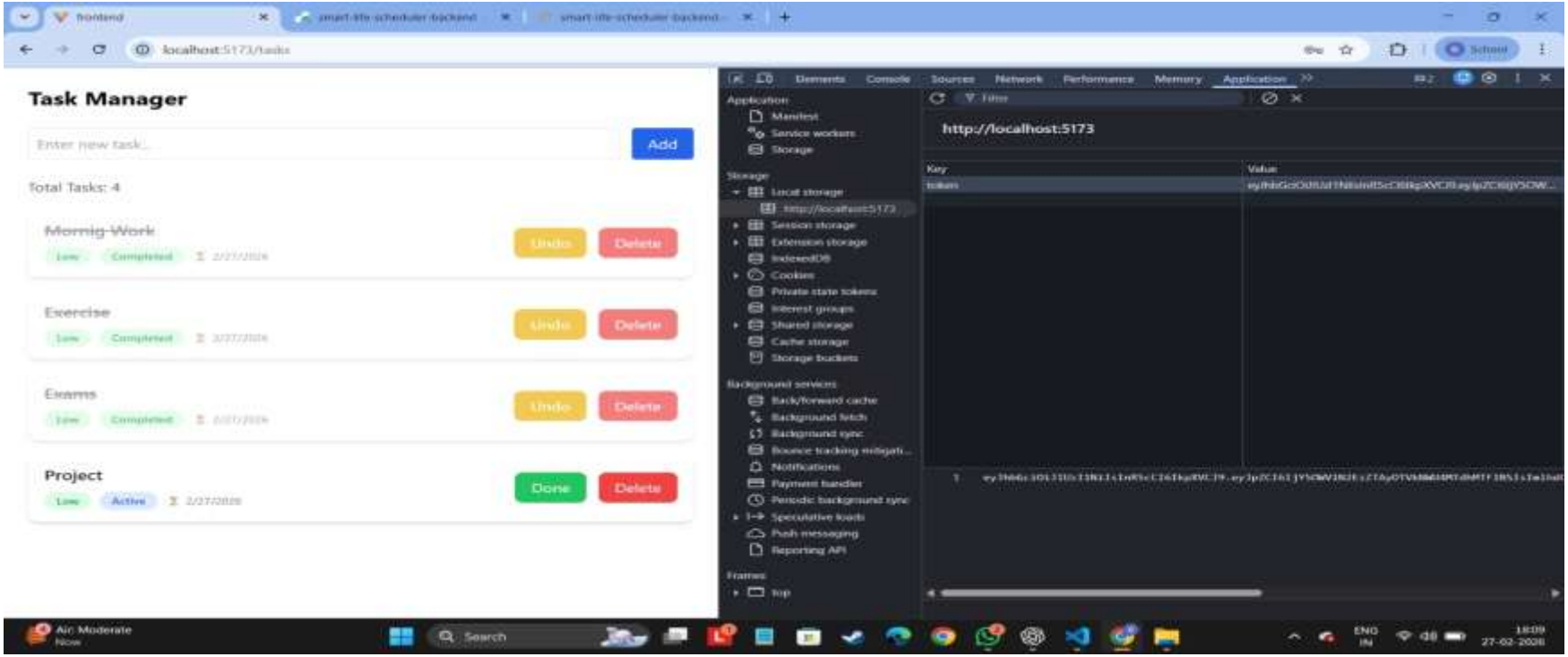
Task manager:
The image reflects completed tasks and pending tasks.



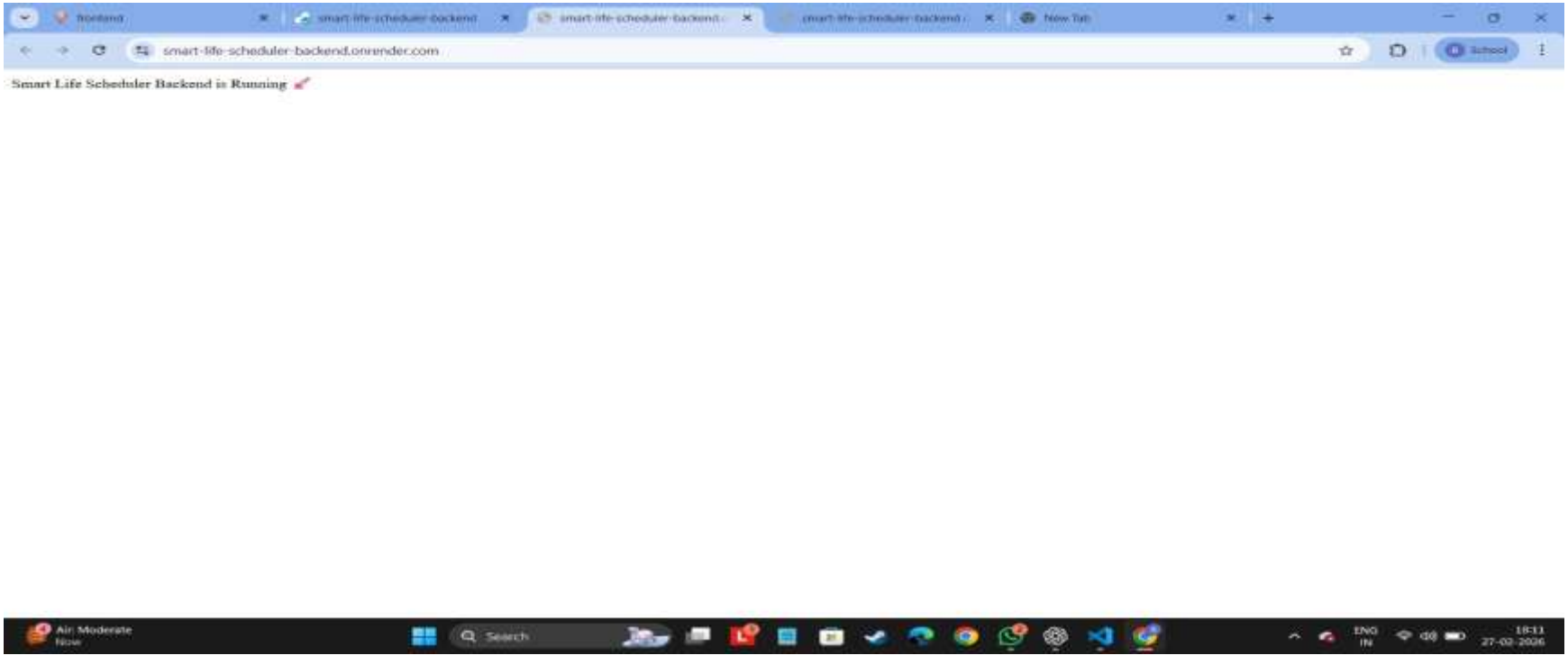
Requests are done using the postman.com.



Network Status working properly in the dev tools of the browser.



Local storage is working and generating the JWT_SECRET token.



Deployed Backend link: https://smart-life-scheduler-backend.onrender.com/

Sustainability

The proposed Smart Daily Scheduler with Integrated Health Monitoring and Automated Reporting is highly sustainable across environmental, economic, and social dimensions. By replacing paper-based planners, manual health logs, and physical reports, the system significantly reduces paper consumption and minimizes the environmental footprint associated with traditional scheduling and health-tracking methods. The solution is developed using open-source and resource-efficient technologies such as Flutter/React Native, Firebase/MongoDB, and Python-based machine learning, making it cost-effective to build, deploy, and maintain over time. Its modular architecture—combining task management, health tracking, ML-driven insights, and automated reporting—ensures long-term technological sustainability and allows easy integration of new features or additional health metrics without major redesign.

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